

Simulation Results

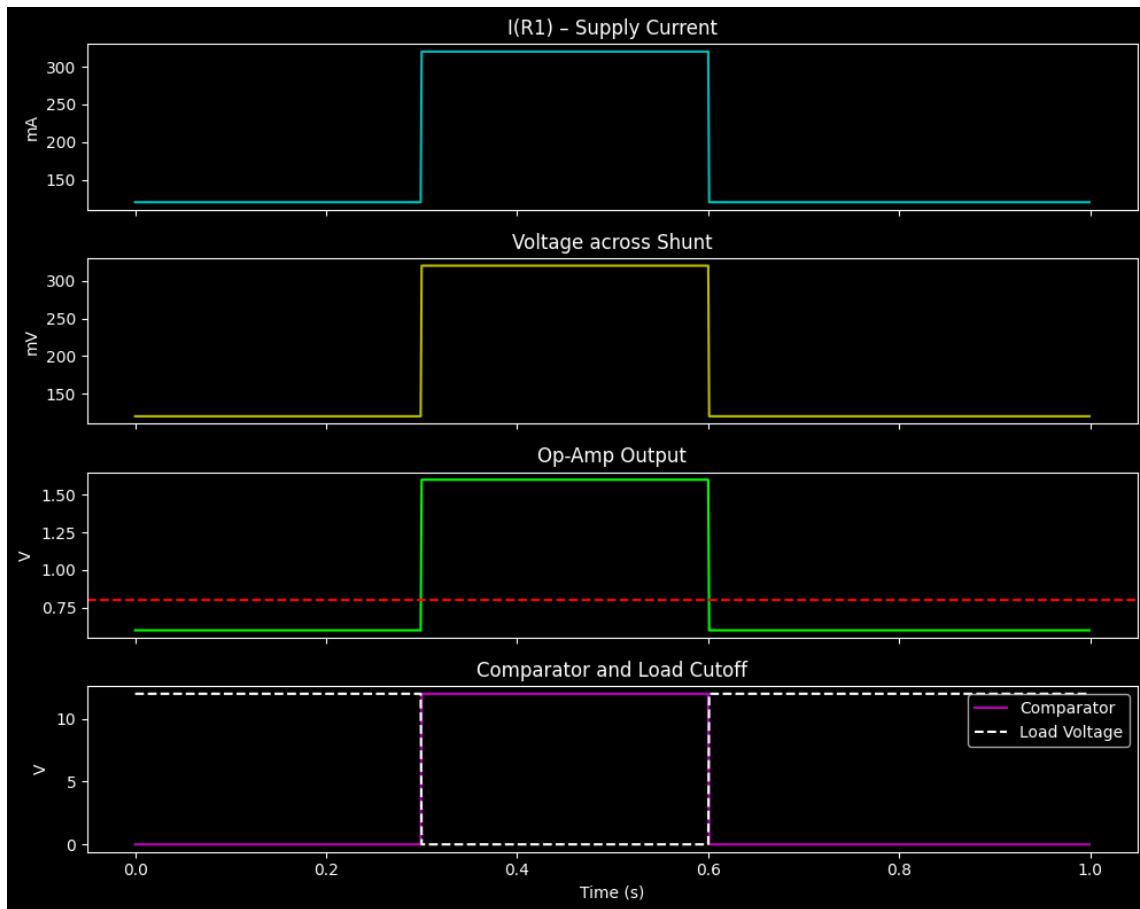


Figure 2. Simulated waveform showing current, shunt voltage, op-amp output, and theft detection cutoff.

Result Explanation:

- The top trace represents the **supply current** through the sensing resistor (R1). Between **0.3 s and 0.6 s**, a sharp rise in current indicates the simulated theft event.
- The second trace shows the **voltage across the shunt resistor**, which increases during theft due to higher current flow.
- The third trace is the **op-amp output**, which amplifies the sensed voltage. When this signal crosses the **reference voltage (0.8 V)**, it triggers the comparator.
- The fourth trace shows the **comparator output** going HIGH (12 V) during theft detection, while the **load voltage** simultaneously drops to **0 V**, indicating automatic cutoff by the MOSFET switch.

This confirms the circuit successfully detects theft, activates an alarm, and isolates the load.

Conclusion

The Smart Energy Theft Detection and Auto Cutoff system was successfully simulated using eSim (Ngspice).

The results verify that the circuit effectively senses abnormal current due to theft, triggers an

alert, and disconnects the power supply automatically.

This design can be extended for real-time smart meter integration and power monitoring applications.

Software Used: eSim (FOSSEE IIT Bombay)

Tools Used: KiCad for schematic, Ngspice for simulation

Simulation Time: 1 s transient analysis

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Github Link: <https://github.com/krupawaani-cmd/Smart-Energy-Theft-Detection-and-Automatic-Cutoff-using-eSim/tree/main>